BSAI FINAL PROJECT
Requirements Specification

Detectra AI

Project Advisor

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Software Requirements Specification

Version 1.0

Detectra AI

Advisor: Usman Aamer

Group ACT-6

Member Name	Primary Responsibility
Abdul Rehman	Implementing backend services, model inference engines, and secure video processing pipelines.
Eman Sarfraz	Managing full project docs and co-developing transformer-based alignment module.
Ahmad Yasin	Training custom logo/motion models, preprocessing with OpenCV/Librosa, and real-time sync engine.

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Revision History

Name	Date	Reason For Changes	Version

Previous Phases Feedback

Idea Defence Feedback (Screenshot)

Following is the feedback regarding your Idea Defence:

Sr. #	**ACT-6**
Project Title	**DETECTRA AI**

They explained well. Web app for vide analysis, fine tuning. Video and prompt will be provided to detect various objects through APIs for logo, object, motion and audio.

Prepare your Proposal Document in the light of comments provided. Last date to submit the Proposal Document is **Wednesday, September 17, 2025** not later than **04:00 PM**. The proposal document should be prepared using the template available at http://smrms.ucp.edu.pk/All_Phases_Templates.html. Submit the hard copy to Project Office in F-304 (Main Building).

When submitting the hard copy of the proposal document, use the duly signed Group Allocation Yellow Page (available at http://smrms.ucp.edu.pk/Forms.html) as the first page of the submission document. Also get the proposal tape bind before submission.

Abstract

Detectra AI is a unified web application designed for multimodal video analysis, providing an end-to-end solution for extracting synchronized insights from raw video data. Unlike existing systems that rely on third-party APIs, **Detectra AI** integrates object detection, logo recognition, motion/activity recognition, and audio transcription (including environmental sounds and speech) into a single platform. Using a transformer-based fusion engine, the system synchronizes outputs from multiple modalities into a unified, timeline-based dashboard, enabling deeper contextual understanding and real-time analysis. This project addresses key challenges in current video analysis tools, including limitations in contextual integration, privacy concerns, and scalability issues associated with API dependencies. By offering a self-contained, API-free framework, **Detectra AI** ensures enhanced privacy, customization, and scalability, making it ideal for applications in surveillance, sports analytics, media monitoring, and other domains. The project covers a broad range of technical knowledge, including deep learning (CNNs, Transformers), computer vision (YOLO, DETR, Vision Transformers), natural language processing (speech-to-text), and web development (Flask, Next.js), culminating in an interactive, timeline-based dashboard that delivers actionable insights across various industries.

1. Introduction and Background

Detectra AI is an innovative project aimed at revolutionizing video analysis through multimodal AI integration. This section provides the context, domain knowledge, and motivation for the project, highlighting the need for a self-sufficient platform in an era of exponential video data growth.

1.1 Product (Problem Statement)

Existing video analysis systems rely heavily on third-party APIs, which limits customization and raises privacy concerns. These systems process different modalities (visual, audio, text) separately, failing to capture holistic context and lacking unified reporting mechanisms. The core problem is the lack of a **self-sufficient, multimodal video intelligence platform**.

1.2 Background

Video data growth has been exponential, but analysis tools are fragmented, often requiring separate APIs for visual objects, audio cues, or text. This modularity hinders real-time contextual understanding and creates dependence on external services, leading to issues in data privacy, scalability, and integration. Recent multimodal AI advancements (e.g., CLIP, GPT-4V) show promise but often lack deployment flexibility. Detectra AI is designed to address this by offering a holistic, integrated framework that processes video, audio, and language modalities together in a self-contained manner.

1.3 Scope

The scope of this SRS covers the development of a **unified web application** called Detectra AI for **end-to-end multimodal video analysis**. This includes all required modules for object detection, logo recognition, motion/activity recognition, audio transcription, a **transformer-based fusion engine**, and a final **interactive, timeline-based dashboard**.

1.4 Objective(s)/Aim(s)/Target(s)

- To develop a **web-based application** for video analysis.
- To implement **object detection**, **logo recognition**, **human motion/action recognition**, and **audio analysis** (environmental & speech-to-text).
- To design a **transformer-based multimodal fusion engine** to align video and audio outputs contextually.
- To provide an **interactive dashboard** for timeline-based reports and visual analytics.
- To demonstrate system **autonomy without reliance on external APIs**.

1.5 Challenges

- Designing an efficient fusion model for aligning multimodal data.
- Training/testing custom logo recognition models.
- Real-time synchronization of audio and visual streams.
- Optimizing performance on limited hardware resources.

1.6 Learning Outcomes

- Expertise in **transformer-based multimodal fusion models**.
- Hands-on experience with **video/audio preprocessing and analysis**.
- Proficiency in deploying **end-to-end AI-powered web applications**.
- Understanding of designing **privacy-preserving, API-free systems**.

1.7 Nature of End Product

The end product is a **web-based software package** that applies advanced AI models for unified video intelligence, producing **interactive dashboards** and **timeline reports** for actionable insights.

1.8 Completeness Criteria

S.No.	Criteria	Weightage %
1	Object detection & logo recognition module	20
2	Motion/action recognition module	15
3	Audio analysis (environmental + speech-to-text)	15
4	Multimodal fusion engine (transformer-based)	25
5	Interactive web dashboard (timeline reports & analytics)	15
6	Final integration, testing, & documentation	10

1.9 Business Goals

The software addresses the corporate goal of providing an **API-free, privacy-preserving video analysis tool** and has industry relevance for **surveillance, sports analytics, media monitoring, and smart cities**.

1.10 Related Work/ Literature Survey/ Literature Review

State-of-the-art models (e.g., CLIP, Flamingo, GPT-4V) show progress in joint vision-language understanding but require large infrastructure and still often rely on modular pipelines. Existing video analysis tools are typically limited to single modalities (e.g., YOLO for objects, Whisper for speech) or integrate functionality via third-party APIs. **Detectra AI** differentiates itself by offering an **integrated, API-independent pipeline** with domain-specific adaptability.

1.11 Document Conventions

Italicized text represents project-specific technical terms (e.g., Multimodal Fusion Engine). Acronyms (e.g., SRS, AI, DETR) are defined in Appendix A.

2. Overall Description

2.1 Product Features

Detectra AI's major features are:

- **Multimodal Analysis Pipeline:** End-to-end processing of visual (object, logo, motion) and audio (speech, environmental sound) data from a single video file.
- **Contextual Fusion:** A transformer-based engine to synchronize outputs from multiple modalities.
- **Timeline Reporting:** A dashboard that presents synchronized results chronologically for deep contextual understanding.
- **API-Free Operation:** A self-contained video intelligence framework.

2.2 User Classes and Characteristics

The primary user class is:

- **Analysts/Researchers:** Users from fields like surveillance, sports analytics, media monitoring, and journalism.
    - Characteristics: Technical understanding of video data, require actionable, time-stamped insights, prioritize data privacy and system autonomy.

2.3 Operating Environment

- **Hardware:** Requires a **GPU-enabled workstation** for training/testing the deep learning models.
- **Software/OS:** The application will operate within a standard web browser (for the dashboard) and rely on a backend running Linux/Windows/macOS capable of hosting the development stack.
- **Coexistence:** Must peacefully coexist with standard machine learning libraries and web servers.

2.4 Design and Implementation Constraints

- **Technology Stack:** Must use specific technologies including **PyTorch/TensorFlow**, **Hugging Face** models, **OpenCV**, **Librosa**, **FFmpeg**, and the web framework **Flask/Next.js**.
- **Hardware Limitation:** Optimizing model performance for potentially **limited hardware resources** available for deployment/testing.
- **Model Training:** Requires training/testing **custom logo recognition models**.
- **Data Privacy:** Must ensure compliance with **data privacy** regulations, especially for surveillance use-cases.

2.5 Assumptions and Dependencies

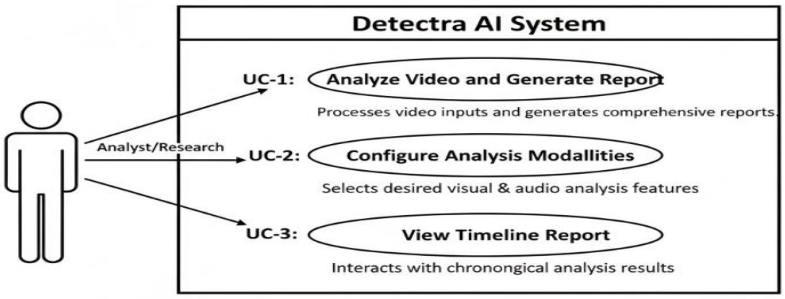
- **Assumptions:**
  - The necessary **GPU hardware** will be accessible for training and testing large models.
  - Standard open-source datasets (OpenLogos, COCO, AudioSet) and custom-curated datasets are available and sufficient for training.
- **Dependencies:**
  - Reliance on the stability and future support of the chosen open-source AI libraries (e.g., YOLO, DETR, Whisper, ViT, wav2vec 2.0).
  - Integration with video and audio pre-processing tools like **OpenCV** and **FFmpeg**.

3. Functional Requirements

UC-0: General Use Case

Identifier	UC-0	
Purpose	To allow the Analyst/Research user to process video inputs and generate a comprehensive report based on the analysis.	
Priority	High	
Pre-conditions	The user must be signed up and have the necessary permissions. A video file must be selected or uploaded for analysis.	
Post-conditions	A comprehensive analysis report is generated and available for viewing and download. The analysis status is updated.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User selects the 'Analyze Video and Generate Report' feature and either uploads a new video or selects one from storage.	System displays video analysis configuration options
2	User confirms the video file and applies the desired analysis modalities (visual, audio, specific features).	System validates the file format and checks the configuration settings.
3	User initiates the analysis process by clicking 'Start Analysis'.	System queues the video for processing, displays a progress indicator, and sends a confirmation message.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	(Analysis Fails) The video file is corrupted, or an internal error occurs during processing.	System stops the analysis, logs the error, and displays an error message to the user: "Video analysis failed. Please try again."
2	(Invalid File) User attempts to upload a non-supported file type (e.g., .doc instead of .mp4).	System rejects the upload and displays an error message: "Invalid file format. Please upload a supported video file."
3	(Cancel Analysis) User cancels the analysis while it is in progress.	System immediately stops processing, cancels the job, and displays a confirmation: "Analysis successfully cancelled."

Table 1: UC-1

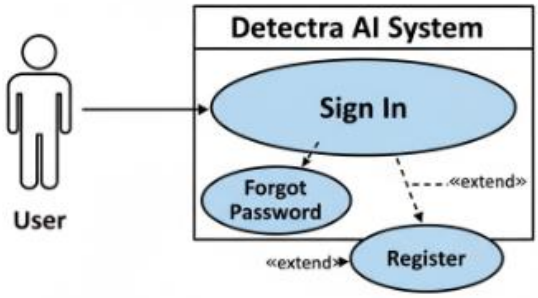


(2.60) General Use-Case Diagram

3.1 UC-1: Sign In

Identifier	UC-1	
Purpose	To allow users to securely sign into the Detectra AI platform to access video analysis features.	
Priority	High	
Pre-conditions	User has a registered account; internet connection is active.	
Post-conditions	User is authenticated and redirected to the dashboard with access to personalized features.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User navigates to the login page.	System displays login form with email and password fields.
2	User enters credentials and clicks 'Sign In'.	System validates credentials against the database.
3	(If valid) User is authenticated.	System redirects user to the dashboard with a welcome message.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	User enters incorrect credentials.	System displays an error message: "Invalid email or password."
2	User selects 'Forgot Password'.	System sends a password reset link to the user's email.
3	User requests a new account.	System redirects to registration page.

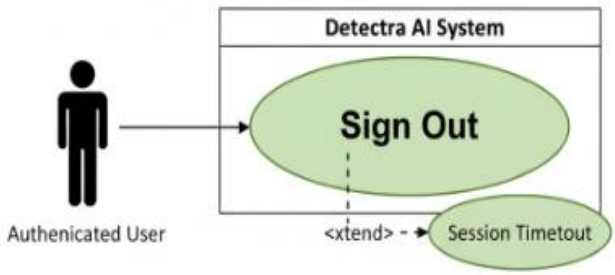
Table 2: UC-1



(2.61) Sign-In

3.2 UC-2: Sign Out

Identifier	UC-2	
Purpose	To enable users to securely sign out of the Detectra AI platform.	
Priority	Medium	
Pre-conditions	User is currently signed in.	
Post-conditions	User session is terminated, and they are logged out.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User clicks 'Sign Out' on the dashboard.	System displays a confirmation prompt.
2	User confirms sign-out.	System terminates the session and clears credentials.
3	User is logged out.	System redirects to the login page with a logout message.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	User cancels sign-out.	System returns to the dashboard.
2	Session timeout occurs.	System automatically logs out and prompts re-login.

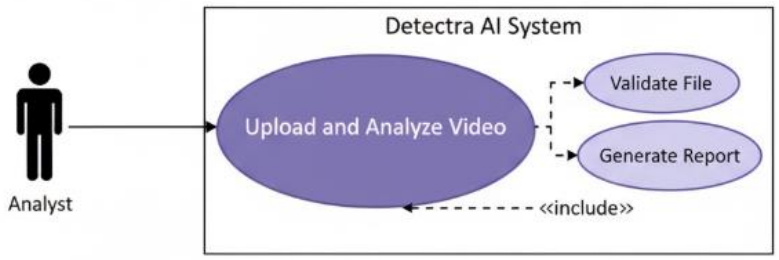


(2.71) Sign-Out

2.8 UC-3: Upload and Analyze Video

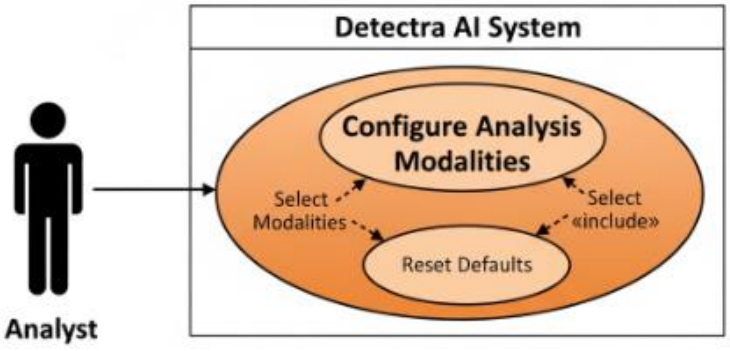
Identifier	UC-3	
Purpose	To allow users to upload and analyze a video file using the Detectra AI platform.	
Priority	High	
Pre-conditions	User is signed in; video file meets format requirements (MP4, AVI).	
Post-conditions	Analysis report is generated and available for viewing.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User navigates to the 'Upload Video' page.	System displays a file upload interface.
2	User selects and uploads a video file.	System validates file format and size, then processes it.
3	Analysis completes.	System generates a timeline-based report and notifies user.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	User uploads an unsupported format.	System displays: "Unsupported file format. Please use MP4 or AVI."
2	Upload fails due to size limit.	System shows: "File exceeds 500MB limit. Try a smaller file."
3	User cancels upload.	System clears the upload queue and returns to the page

(2.81) Analyze and Upload Video



2.9 UC-4: Configure Analysis Modalities

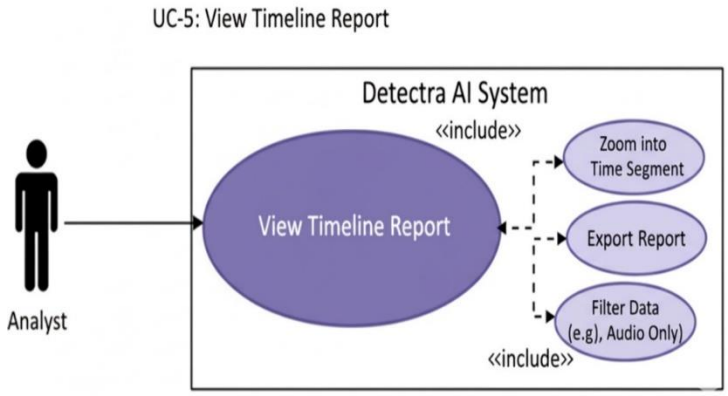
Identifier	UC-4	
Purpose	To enable users to configure analysis modalities (e.g., object detection, audio transcription) before processing a video. a video is uploaded.	
Priority	Medium	
Pre-conditions	User is signed in; a video is uploaded.	
Post-conditions	Analysis is customized based on selected modalities.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User accesses the 'Configure Analysis' page.	System displays toggle options for modalities (e.g., object, audio).
2	User selects desired modalities (e.g., object detection on).	System updates settings and saves configuration.
3	User initiates analysis.	System processes video with selected modalities.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	User deselects all modalities.	System prompts: "Select at least one modality."
2	User resets to default settings.	System reverts to all modalities enabled.



(2.91) Configure Analysis Modalities

2.10 UC-5: View Timeline Report

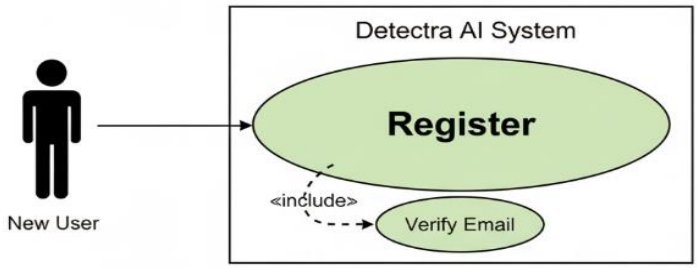
Identifier	UC-5	
Purpose	To allow users to view and interact with the timeline-based analysis report on the dashboard.	
Priority	High	
Pre-conditions	User is signed in; analysis report is generated.	
Post-conditions	User gains insights from the visualized data.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User navigates to the 'Dashboard'.	System displays the timeline report with synchronized events.
2	User zooms into a specific time segment.	System updates the view with detailed event data.
3	User exports the report	System generates a PDF/CSV file for download.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	User filters by modality (e.g., audio only).	System refreshes display with filtered data.
2	Report generation fails.	System notifies: "Export failed. Try again."



(2.10.1) View Timeine Report

2.11 UC-6: Register

Identifier	UC-6	
Purpose	To enable users to register a new account for accessing Detectra AI.	
Priority	Medium	
Pre-conditions	User has no existing account; internet connection is active.	
Post-conditions	User account is created, and they can log in.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User navigates to the 'Register' page.	System displays a registration form (email, password, name).
2	User fills and submits the form.	System validates data and sends confirmation email.
3	User verifies email.	System activates the account and redirects to login.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	User enters duplicate email.	System shows: "Email already registered."
2	User skips email verification.	System locks account until verification.



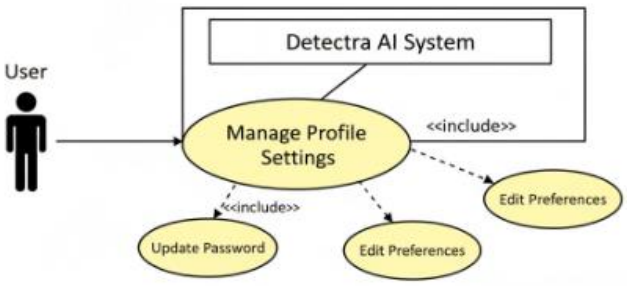
(2.11.1) Register

2.12 UC-7: Manage Profile Settings

Identifier	UC-7
Purpose	To allow users to manage profile settings (e.g., update password, preferences).
Priority	Low
Pre-conditions	User is signed in.
Post-conditions	User profile is updated with new settings.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User navigates to 'Profile Settings'.	System displays current profile details.
2	User updates password or preferences.	System validates changes and saves them.
3	User saves changes.	System confirms: "Profile updated successfully."

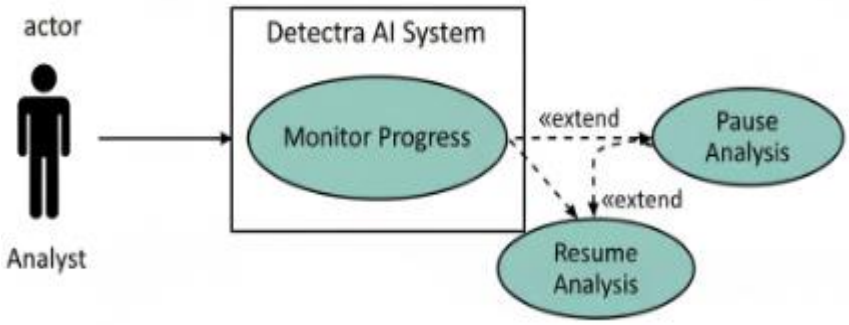
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	User enters mismatched passwords.	System shows: "Passwords do not match."
2	User cancels changes.	System discards updates and returns to profile.



(2.12.1) Manage Profile Settings

3.13 UC-8: Real-Time Progress Feedback

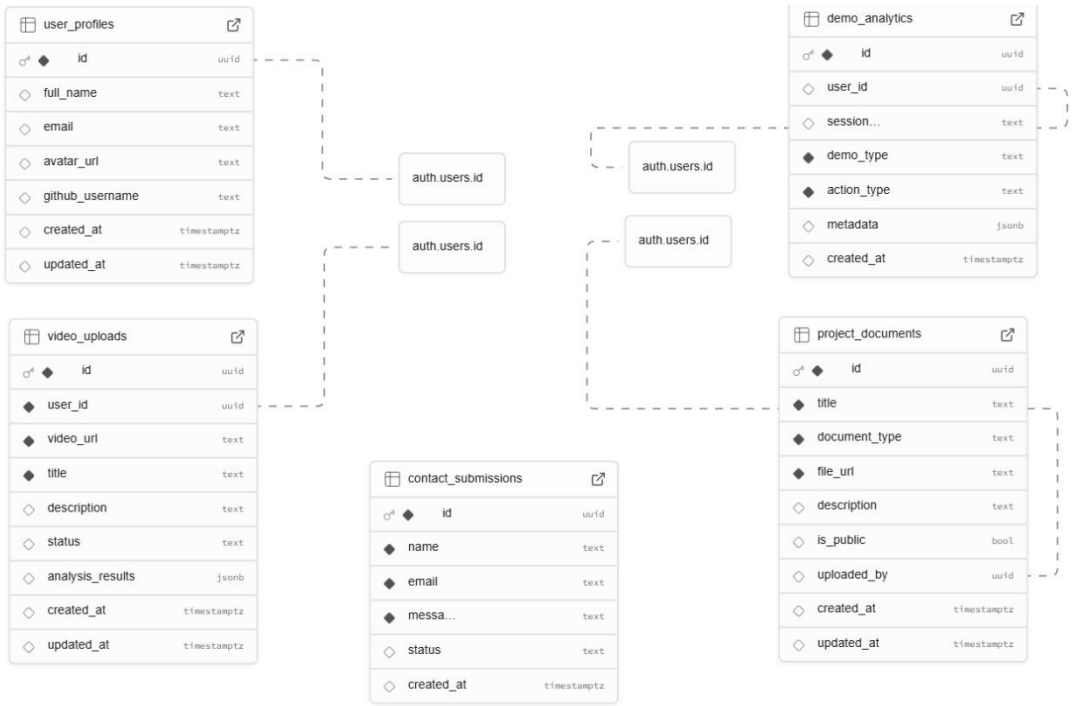
Identifier	UC-8	
Purpose	To provide real-time progress feedback during video analysis.	
Priority	Medium	
Pre-conditions	User is signed in; video analysis is initiated.	
Post-conditions	User receives updates on analysis progress.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	User starts video analysis.	System displays a progress bar (e.g., 0% complete).
2	Analysis progresses.	System updates progress bar (e.g., 50% complete).
3	Analysis completes.	System shows 100% and redirects to report.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	User pauses analysis.	System halts processing and holds progress.
2	Analysis fails midway.	System displays: "Analysis interrupted. Retry?"



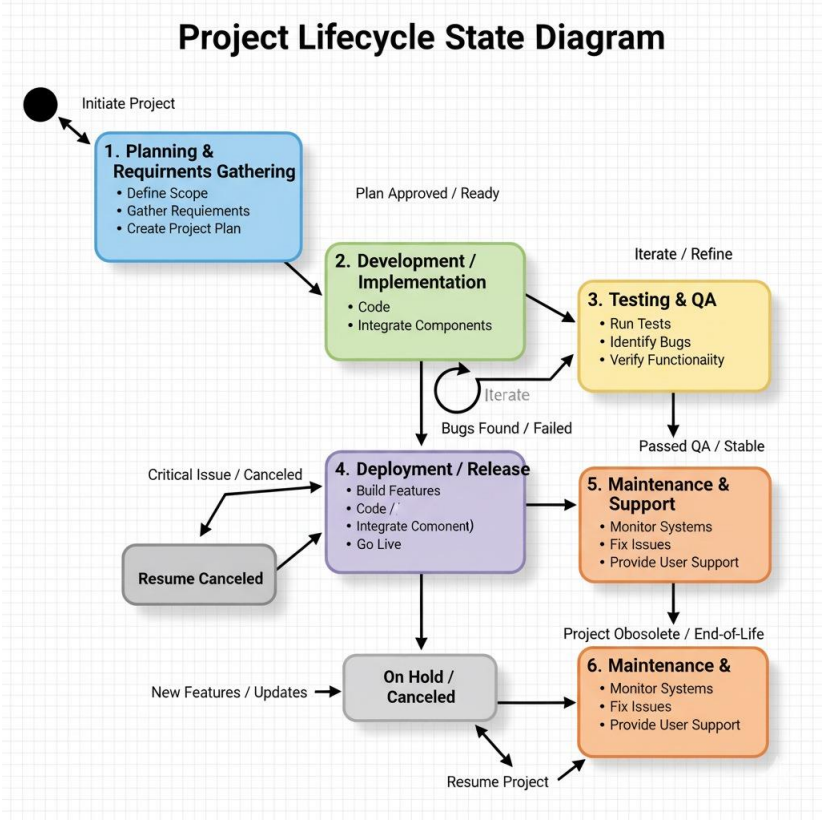
(2.13.1) Progress Feedback

3.9 Requirements Analysis and Modeling

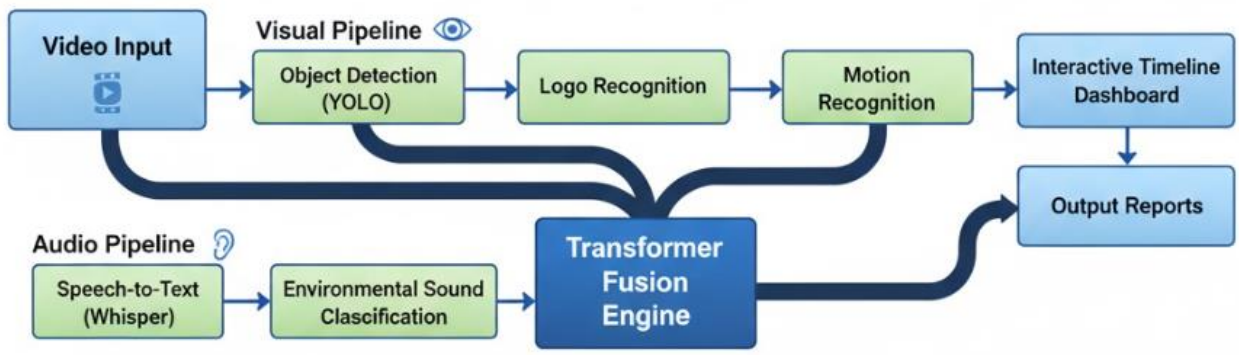
Following are the analysis models: use-case diagram, entity-relationship diagram, sequence diagram, flow chart and state diagram.



(2.91) Entity-Relationship Diagram (Database)

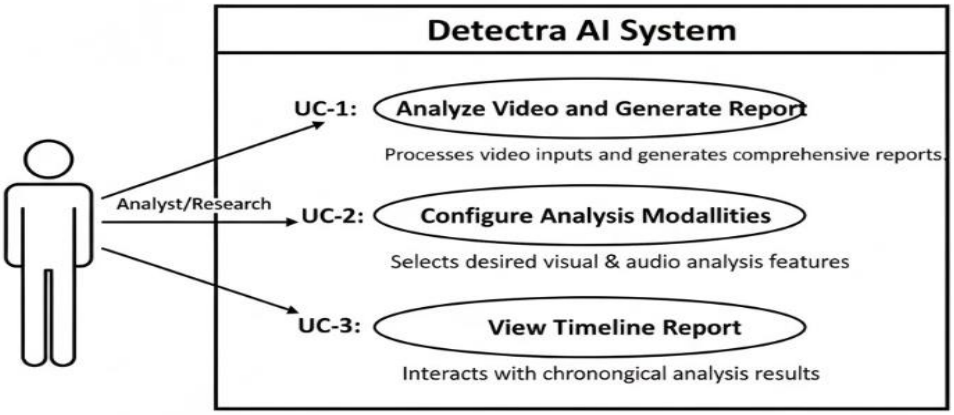


(2.92) State-Diagram



(2.93) Sequence Diagram

(2.94) Flow Chart



(2.95) General Use-Case Diagram

4. Nonfunctional Requirements

4.1 Performance Requirements

- **Processing Time:** The system should process a 1-minute video segment and generate a unified report within a reasonable time limit (e.g., T seconds), prioritizing efficient inference over real-time processing.
- **Scalability:** The framework must be designed to support **large-scale datasets** with **GPU acceleration** to handle future growth.

4.2 Safety Requirements

N/A (Project is a software system; no physical safety concerns beyond standard data center operations).

4.3 Security Requirements

- **Privacy:** Must ensure **data privacy** compliance, especially for sensitive data like surveillance video.
- **Access Control:** User identity **authentication requirements** must be defined for accessing the web application and its reports.

4.4 Additional Software Quality Attributes

- **Reliability:** The core detection and fusion modules must maintain high accuracy for all modalities.
- **Extensibility:** The framework is designed to easily **integrate future modalities** (e.g., biometric recognition, sentiment analysis).
- **Maintainability:** The use of well-established libraries and modular design should ensure high code maintainability.

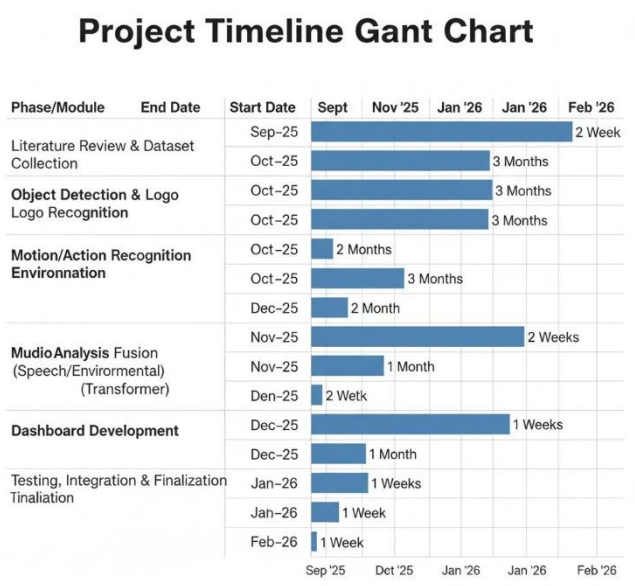
5. Other Requirements

N/A

6. Revised Project Plan

The project plan is presented as a Gantt chart, showing parallel tasks. The duration is approximately **2 regular semesters**.

Phase/Module	Start Date	End Date	Duration
Literature Review & Dataset Collection	Sept 2025	Oct 2025	2 Months
Object Detection & Logo Recognition	Oct 2025	Dec 2025	3 Months
Motion/Action Recognition	Oct 2025	Dec 2025	3 Months
Audio Analysis (Speech/Environmental)	Nov 2025	Jan 2026	2 Months
Multimodal Fusion Engine (Transformer)	Dec 2025	Jan 2026	1 Month
Dashboard Development	Jan 2026	Feb 2026	2 Weeks
Testing, Integration & Finalization	Feb 2026	Feb 2026	1 Week



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Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: IV & V Report

(Independent verification & validation)

IV & V Resource

Name Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
…					

Table 3: List of non-trivial defects